



ReDI School of
Digital Integration

Internal Demo Day: Data Circle

Fall 2025

DCP Hamburg



We use tech to connect human potential and
opportunity with dignity & humility

Flu Vaccination Predictor Project

By ALPHA GROUP

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 Based on CDC H1N1 2009 Survey Data

Introduction



Seasonal and pandemic flu vaccines are vital public health tools. Despite availability, uptake varies due to beliefs, demographics, and access. This study uses machine learning to understand and predict who gets vaccinated

Objectives:

- ❑ Analyse public attitudes and behaviours toward vaccination
- ❑ Predict uptake of:
 - H1N1 vaccine
 - Seasonal flu vaccine
- ❑ Interpret model outputs to find key influencing factors
- ❑ Offer data-driven recommendations for public health strategy

Modeling Approach

- ❑ Random Forest Classifier:
 - Captures nonlinear patterns
 - Suitable for feature importance
- ❑ Logistic Regression:
 - Interpretable linear model
 - Good for policy recommendations
- ❑ Separate models trained for H1N1 and Seasonal vaccine prediction

Dataset Overview

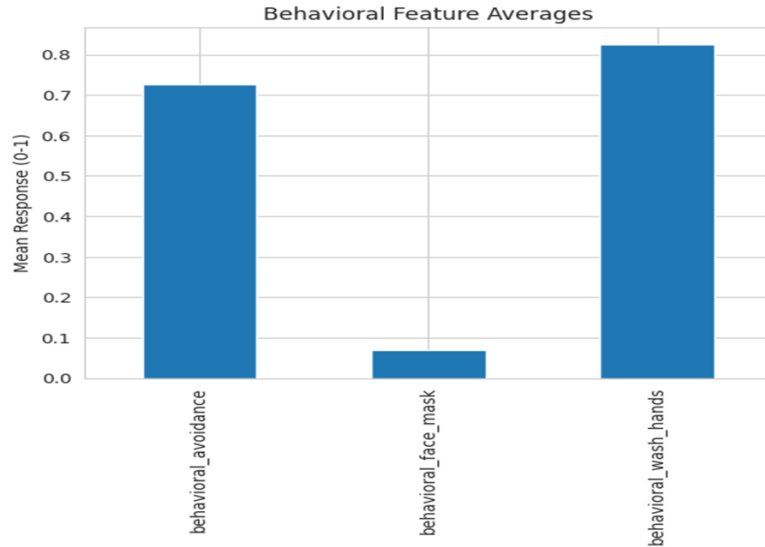
- **Source:** CDC 2009 H1N1 Flu Survey
- **Size:** 26,707 respondents across U.S, 36 features
- **Features:**
 - Demographics (age, race, education)
 - Health behavior (doctor visits, chronic conditions)
 - Opinion (concern about flu, confidence in vaccines)
- **Two binary target variables:**
 - H1N1 Vaccine (Yes/No)
 - Seasonal Vaccine (Yes/No)



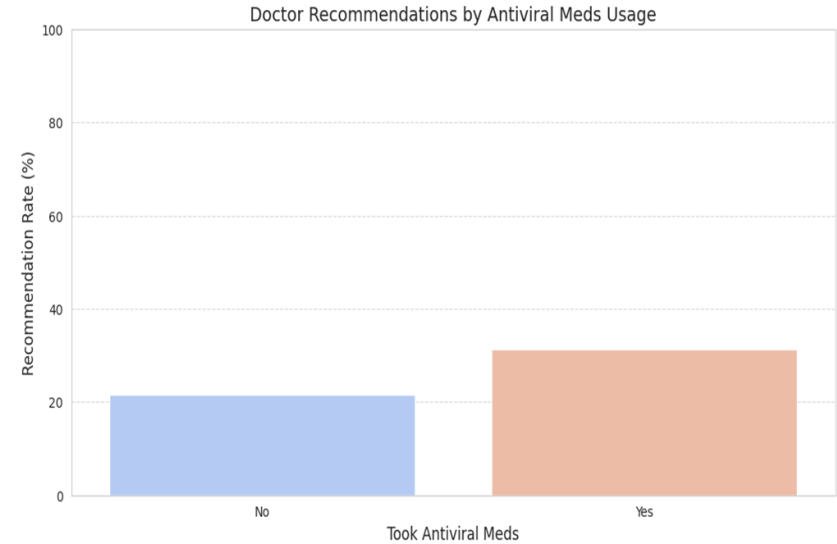
Exploratory Data Analysis (EDA)



Concern about H1N1 Virus = more handwashing



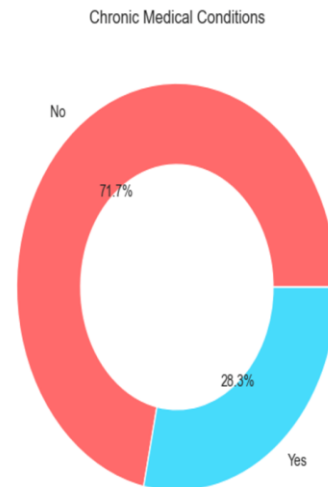
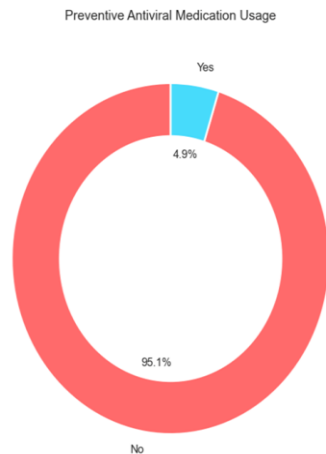
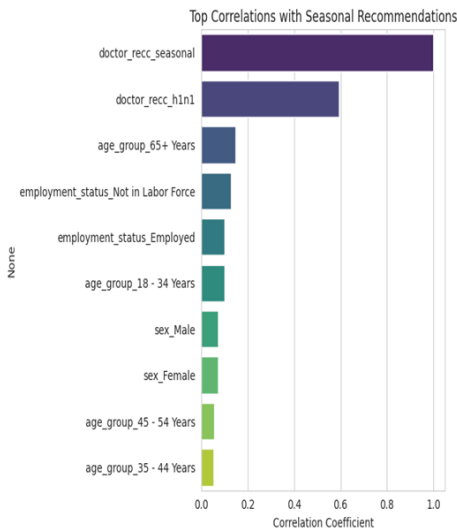
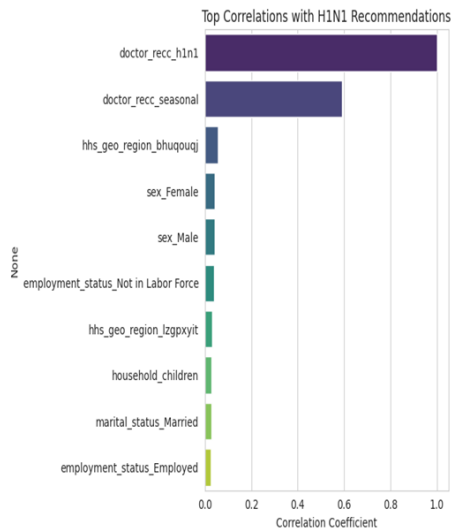
Doctor Recommendations Influence Vaccination



Exploratory Data Analysis (EDA)

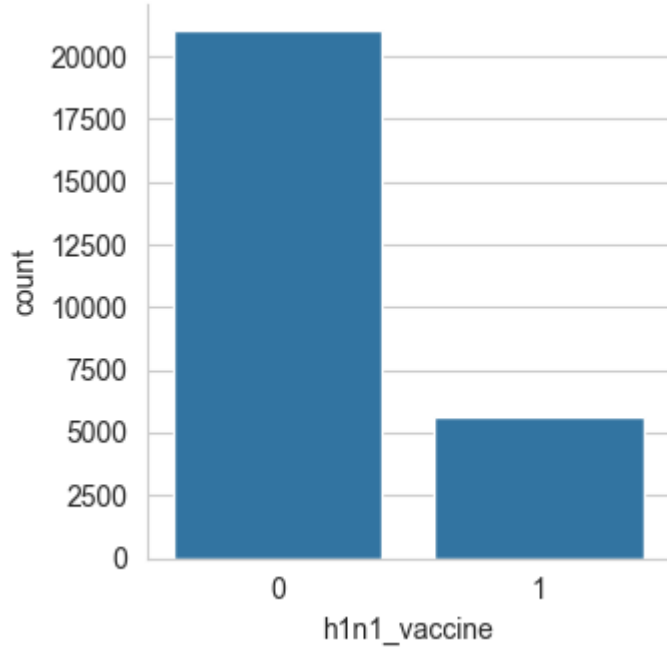
Older adults (65+) more likely to get seasonal flu shots

4.9% Persons takes Preventive Antiviral Medication while, 28.3% Persons has chronic medical conditions

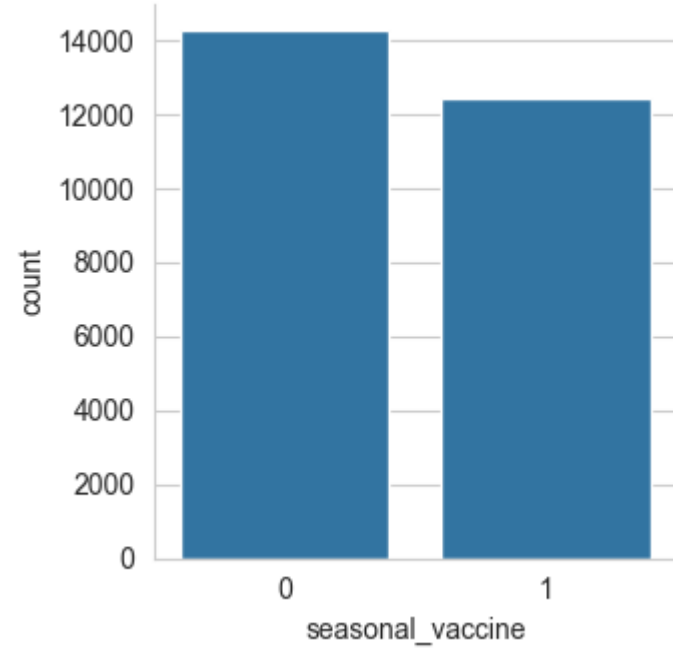


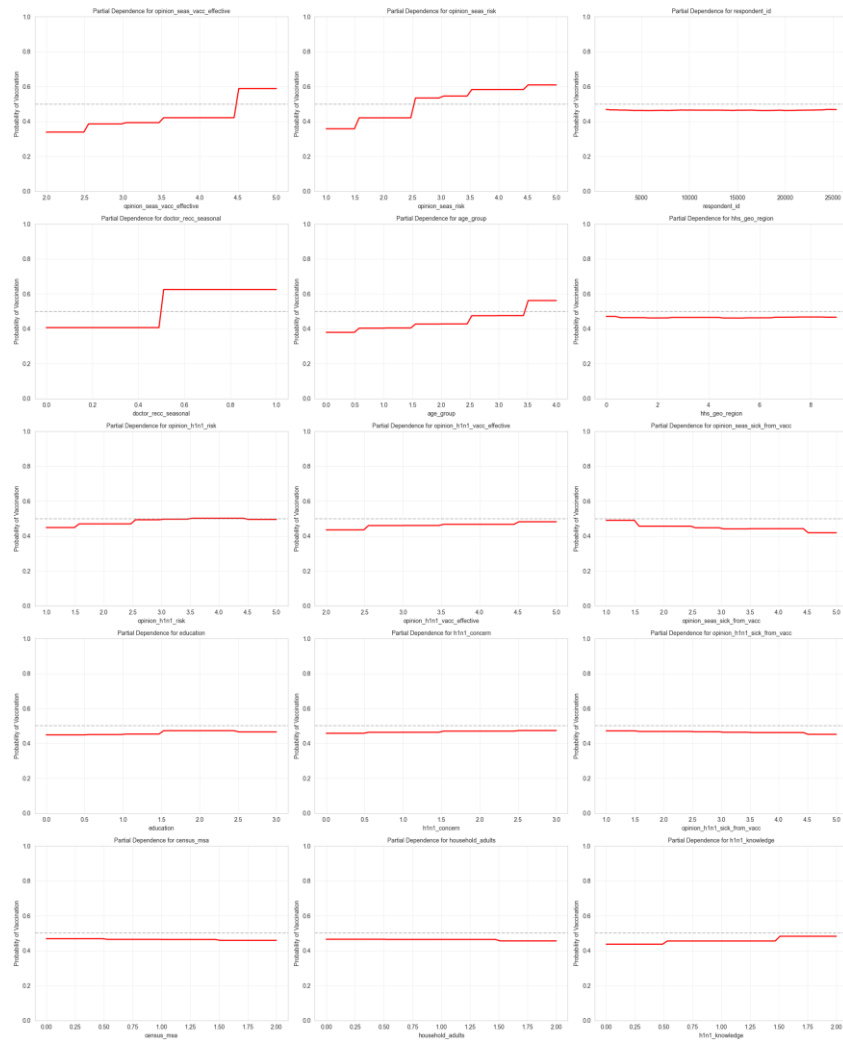
Target Variable Distribution

H1N1 Vaccine: Majority did not take it



Seasonal Vaccine: Higher uptake





Partial Dependence Plots (H1N1 Vaccine)



Key Findings:

•Beliefs and Opinions Matter Most:

- $\uparrow$ *Belief in vaccine effectiveness* $\rightarrow$ $\uparrow$ Likelihood of vaccination
- $\uparrow$ *Flu risk perception* $\rightarrow$ $\uparrow$ Likelihood of vaccination
- $\downarrow$ *Fear of vaccine side effects* $\rightarrow$ $\uparrow$ Vaccination probability

•Doctor's Recommendation:

- When a **doctor recommends** the seasonal flu shot, people are **significantly more likely** to take the H1N1 vaccine.

•Age Factor:

- Older age groups show **higher vaccination likelihood**

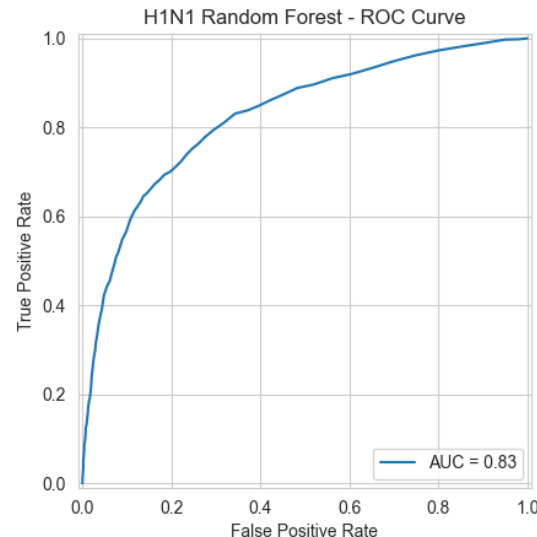
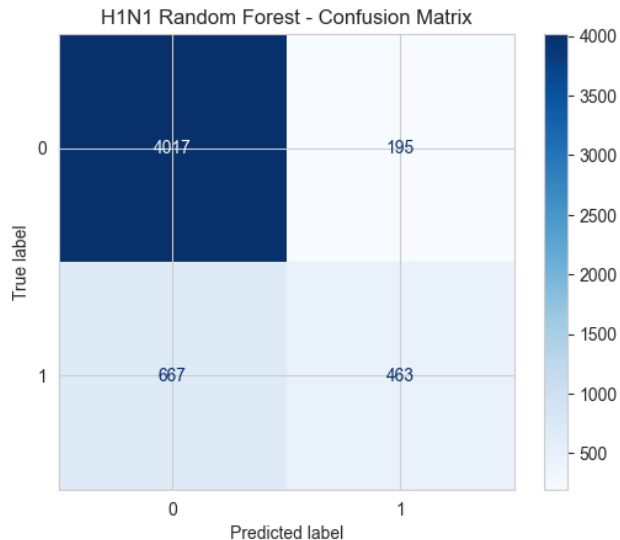
•Education and Concern:

- Higher education and **concern about H1N1** modestly increase uptake.

•Minimal Influence from:

- Demographic features like region, number of adults in household, and respondent ID.
- Lines for these are mostly **flat**, showing weak effect.

H1N1 Vaccine– Random Forest Performance



Accuracy (0.84): The model correctly predicts 84% of cases.

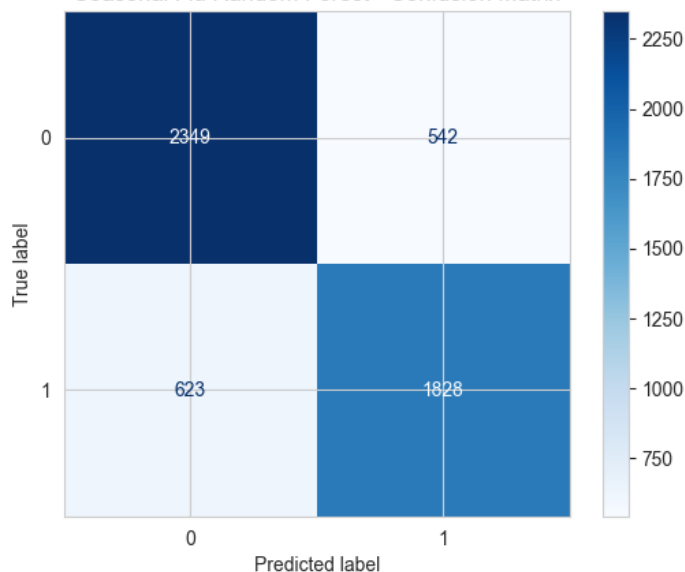
ROC AUC (0.83): Good ability to distinguish between vaccinated and unvaccinated individuals.

F1 Score (0.52): Moderate balance between precision and recall — suggests the model struggles more with imbalanced classes or identifying positives accurately.

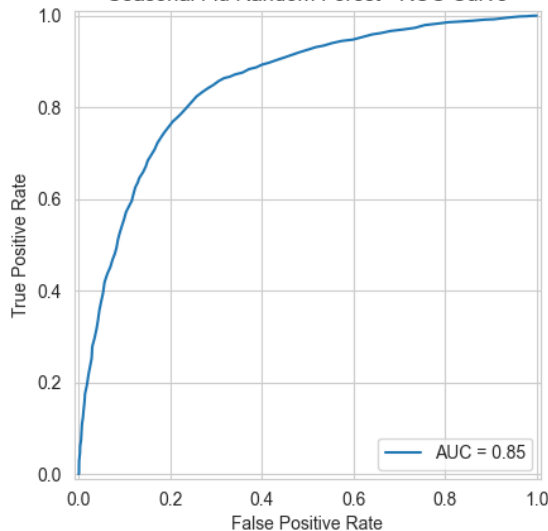
Interpretation: While overall accuracy and AUC are good, the low F1 score suggests the model may be missing many vaccinated individuals (low recall or precision).

Seasonal Flu Vaccine – Random Forest Performance

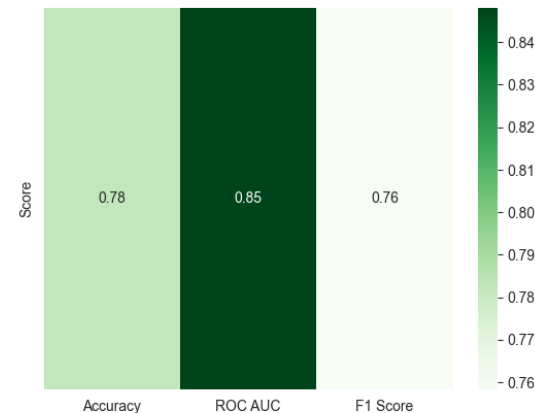
Seasonal Flu Random Forest - Confusion Matrix



Seasonal Flu Random Forest - ROC Curve



Seasonal Flu – Random Forest Metrics



Accuracy (0.78): Slightly lower overall correctness than H1N1.

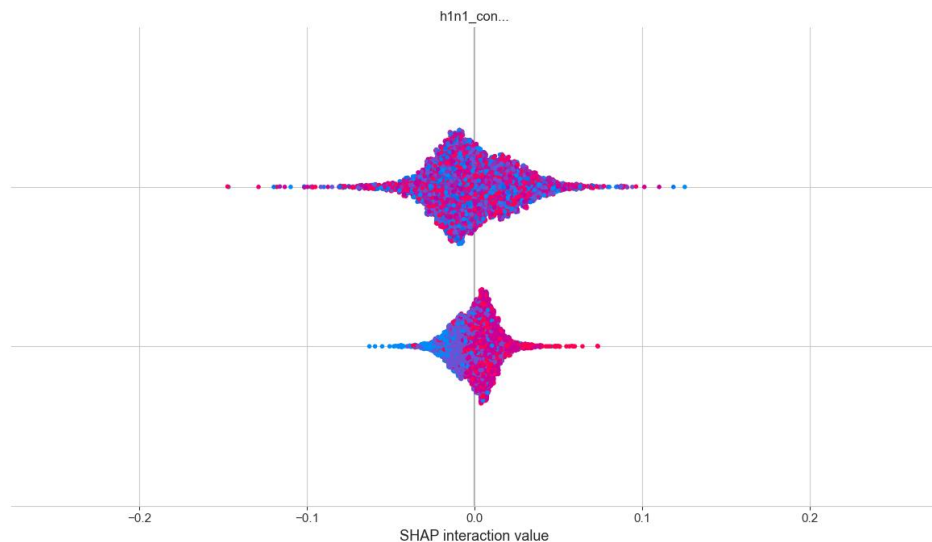
ROC AUC (0.85): Strong ability to rank predictions.

F1 Score (0.76): High — indicates good balance between precision and recall.

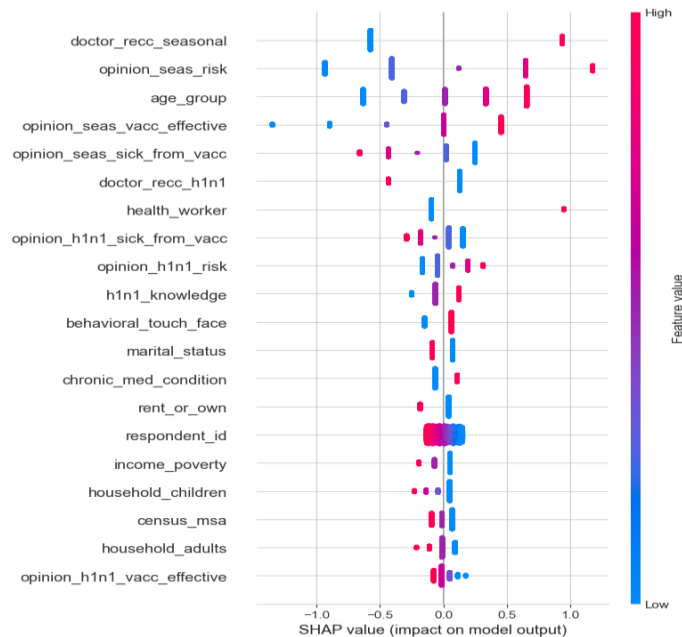
Interpretation: The seasonal flu model performs better in terms of F1 score, meaning it's more effective at identifying vaccinated individuals reliably, despite a slightly lower accuracy.

SHAP – Model Interpretability

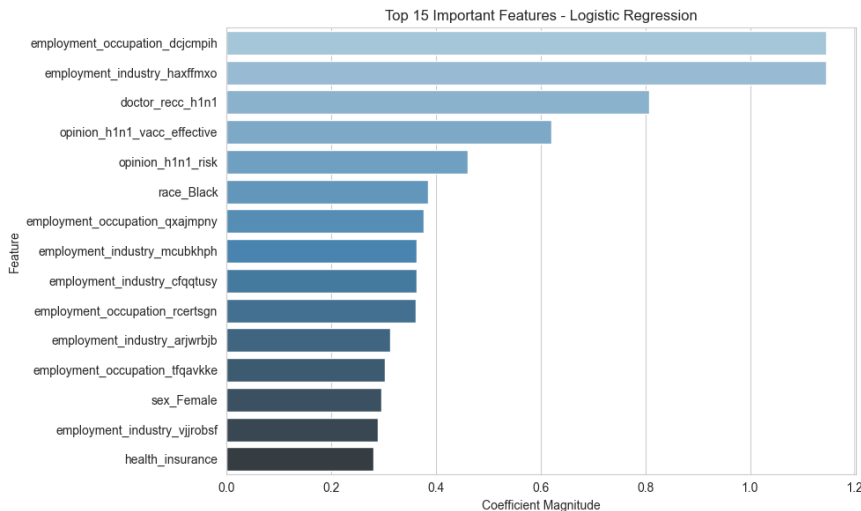
SHAP Summary Plot - H1N1 Vaccine (Random Forest Classifier)



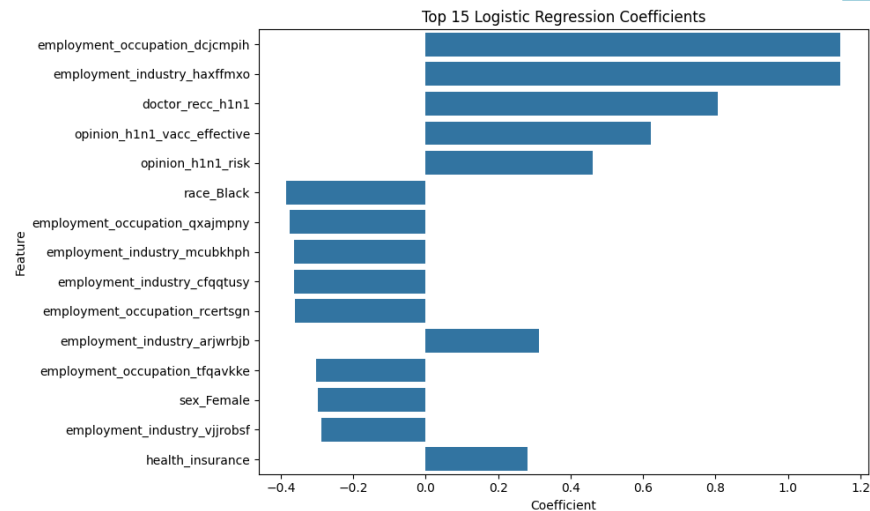
SHAP Summary Plot - SEASONAL Vaccine (Logistic Regression)



Logistic Regression – Feature Weights



This graph highlights that **occupation, medical advice, and personal opinions** are the strongest drivers in H1N1 vaccination behaviour. Demographic factors and insurance also play a role but are less dominant.



This bar chart displays both positive and negative influences on H1N1 vaccination based on a logistic regression model. Key predictors like **occupation, industry, and doctor recommendations** increase the likelihood of vaccination. Positive opinions about vaccine effectiveness and risk also encourage uptake. Conversely, certain demographics and job sectors are linked to lower vaccination rates.

Key Features, Insights & Recommendations

Top Influencing Features

•High impact features common across both SHAP and LR features models:

- Vaccine confidence
- Doctor recommendation
- Risk perception
- Insurance and access to care

Key Insights

- Trust, access, and risk perception drive decisions
- Information campaigns must address safety concerns
- Doctor recommendation is a strong influence
- Younger populations and uninsured less likely to vaccinate

Recommendations

- Strengthen **vaccine communication campaigns**
- Address **vaccine hesitancy** with data-driven messaging
- Focus on groups:
 - Without health insurance
 - With low vaccine trust
 - Younger adults
- Encourage doctors to promote vaccines during visits



Thank you!

